

Diagram illustrating the Tucker decomposition of a 3D tensor T into a core tensor G and three factor matrices $A_1, A_2, \dots, A_r$.

The tensor T is shown as a purple cube, labeled T , with dimensions $\in \mathbb{R}^{n \times m \times p}$.

The decomposition is represented by the equation:

$$T = A_1 G A_2 \dots A_r$$

where:

- $A_1, A_2, \dots, A_r$ are factor matrices (blue vertical bars).
- G is the core tensor (light blue cube).
- $c_1, c_2, \dots, c_r$ are the dimensions of the core tensor G along the first, second, and r -th modes, respectively (green diagonal bars).
- $b_1, b_2, \dots, b_r$ are the dimensions of the factor matrices $A_1, A_2, \dots, A_r$ along the second, third, and r -th modes, respectively (orange horizontal bars).